

NAPOLEON'S FATE



DONT OPEN ON MOBILE :)



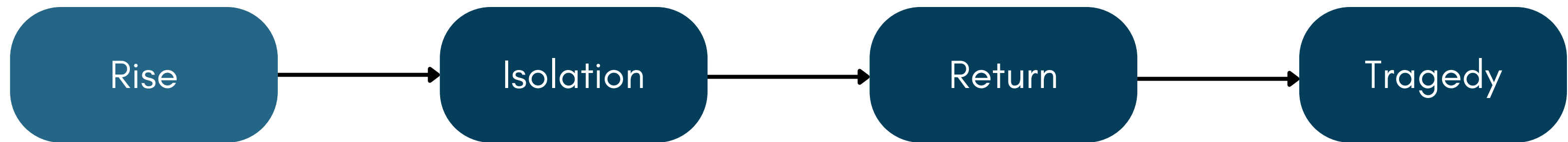
ix-0.github.io/ICG/

THE IDEA

Available in story.md file

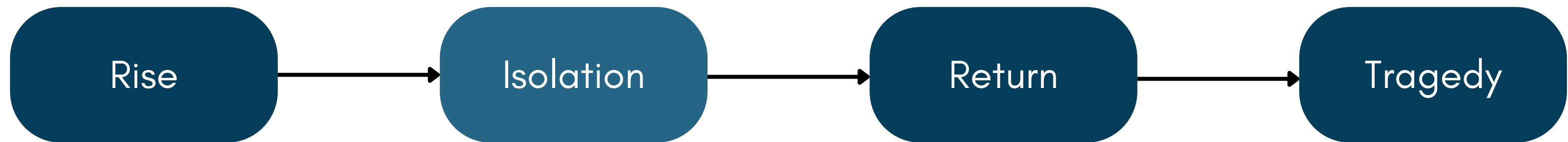


STAGES



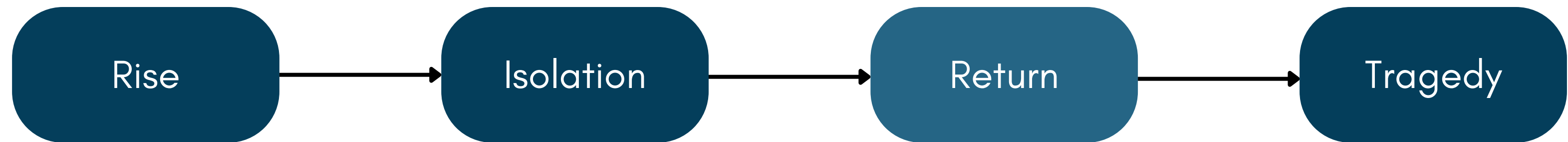
“Rise” represents the rise in power Napoleon had during the french revolution

STAGES



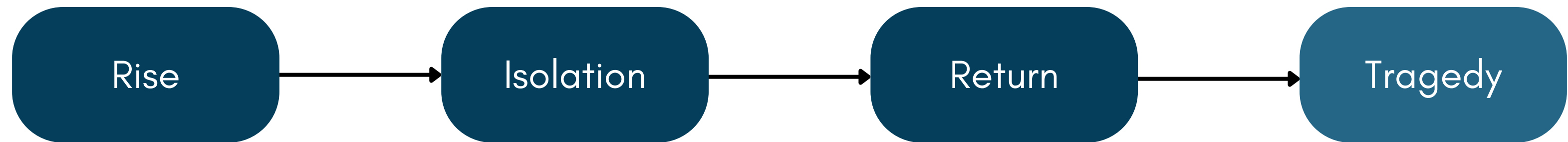
“Isolation” represents the lost of power after many European nations joined forces to exile him

STAGES



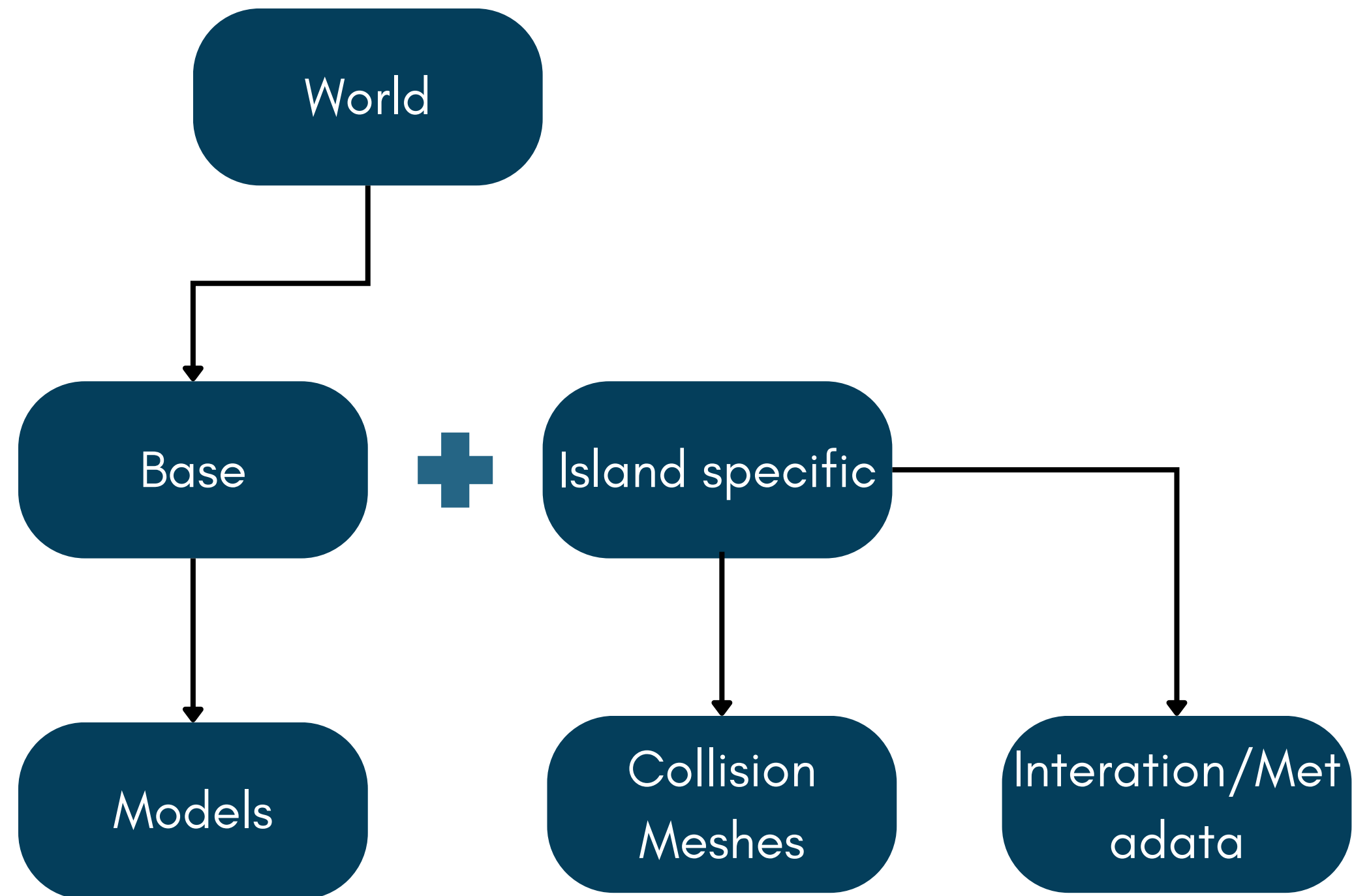
"Return" represents the known 100 day period in which
Napoleon regained its power

STAGES



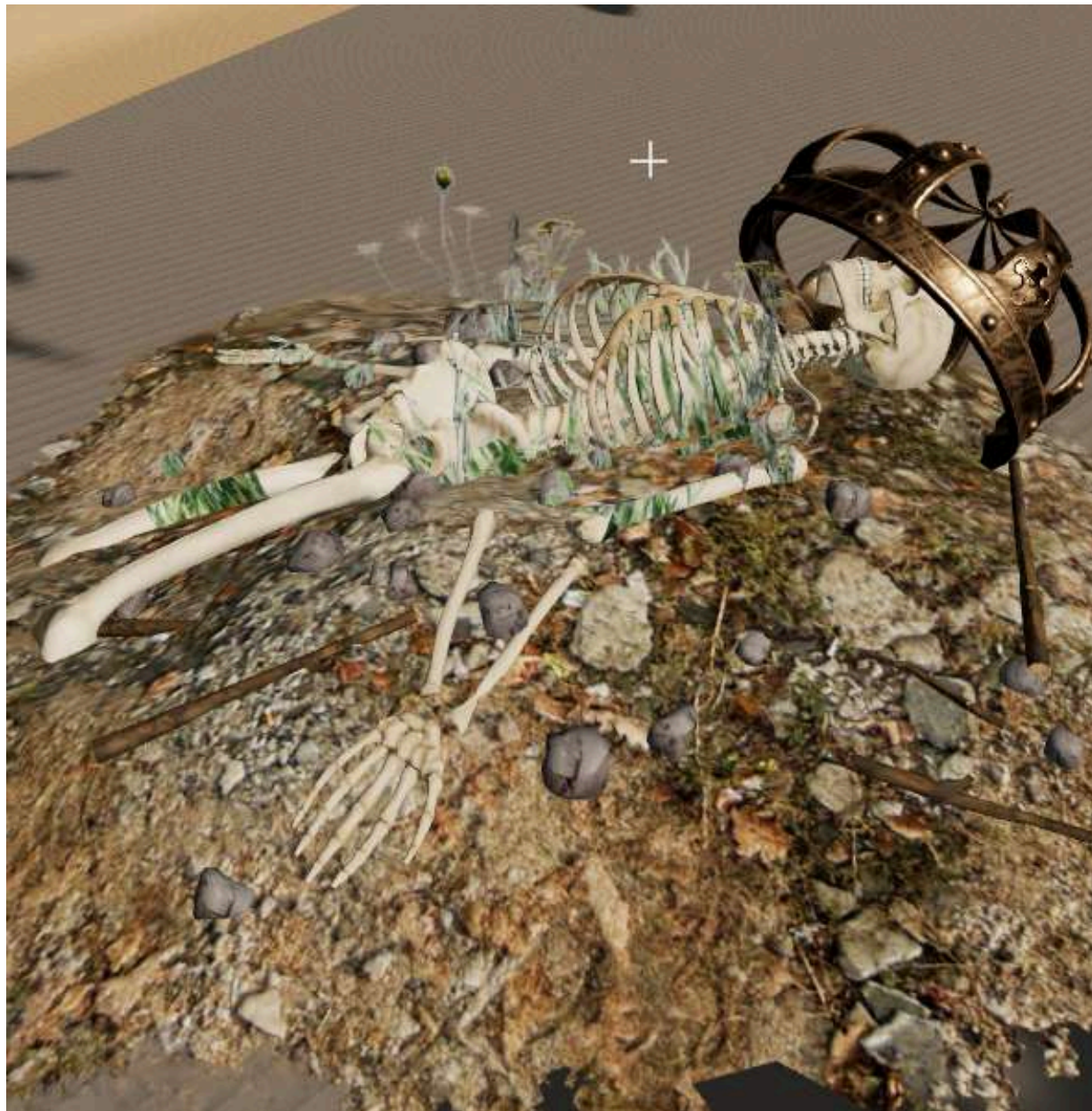
Which ended in his permanent exile after the famous Waterloo battle
which is represented in the “Tragedy”

HIERARCHY



RISE

The power



Really achieving it

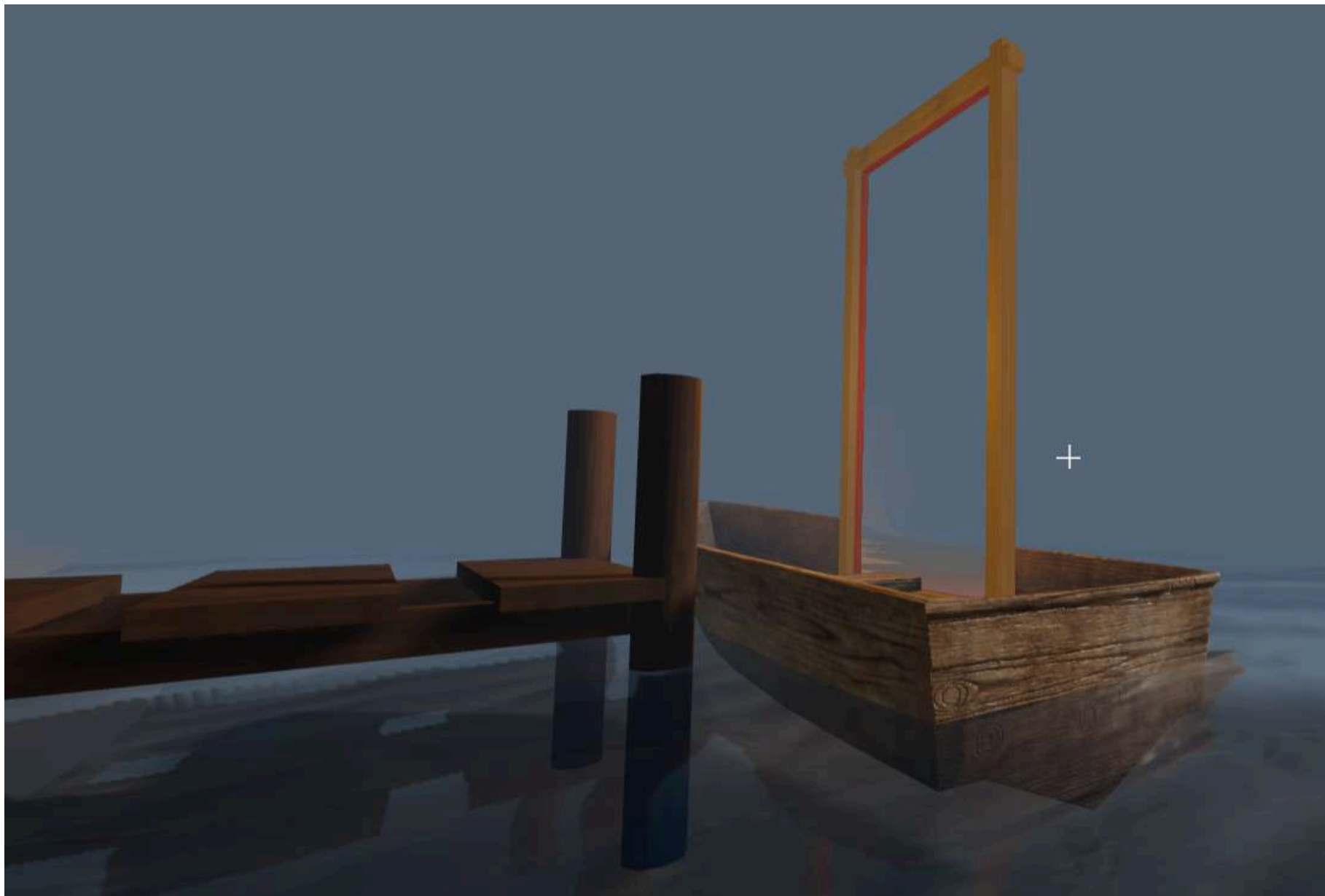


RISE

Transition to isolation



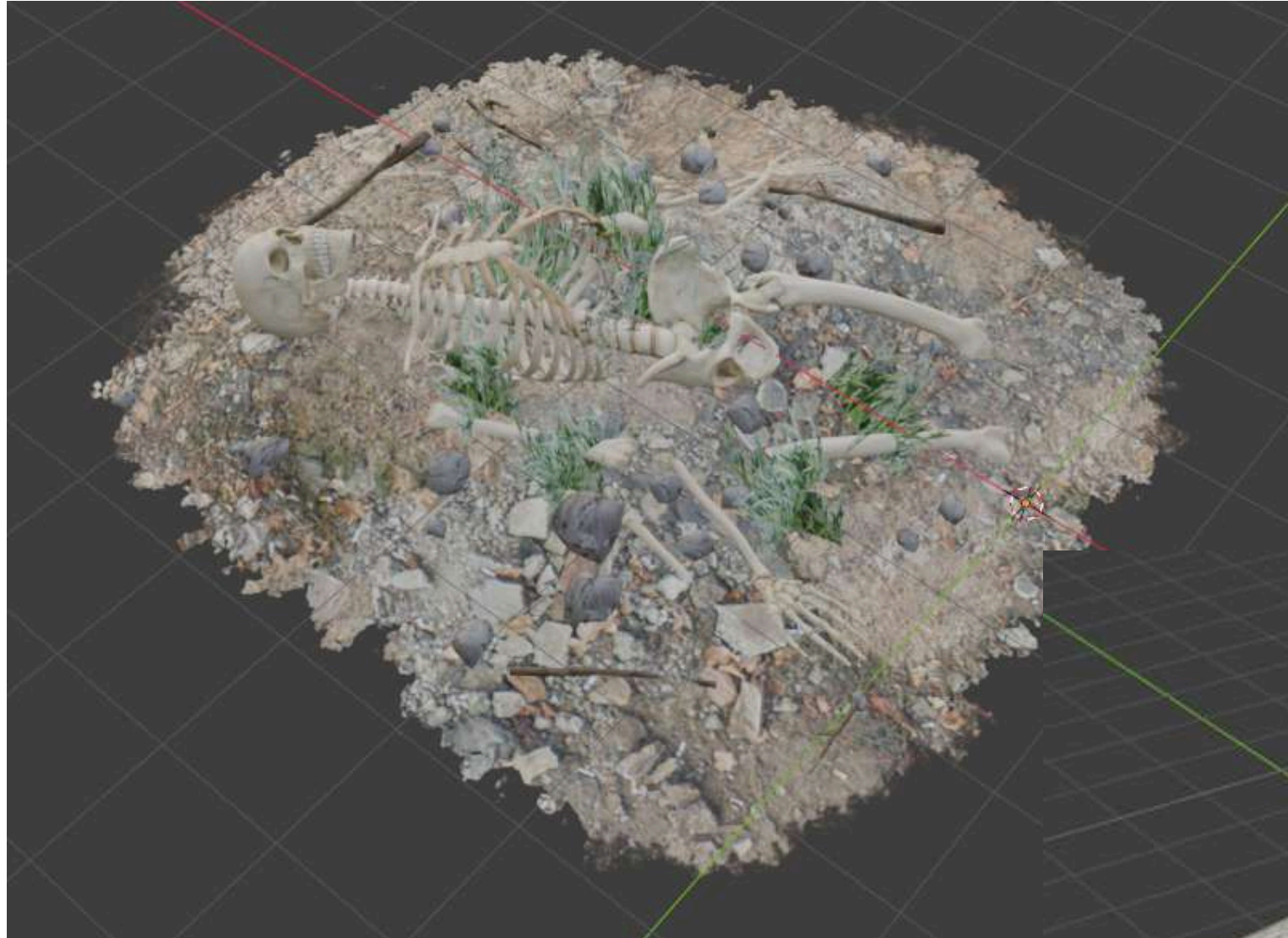
RISE



The Portal

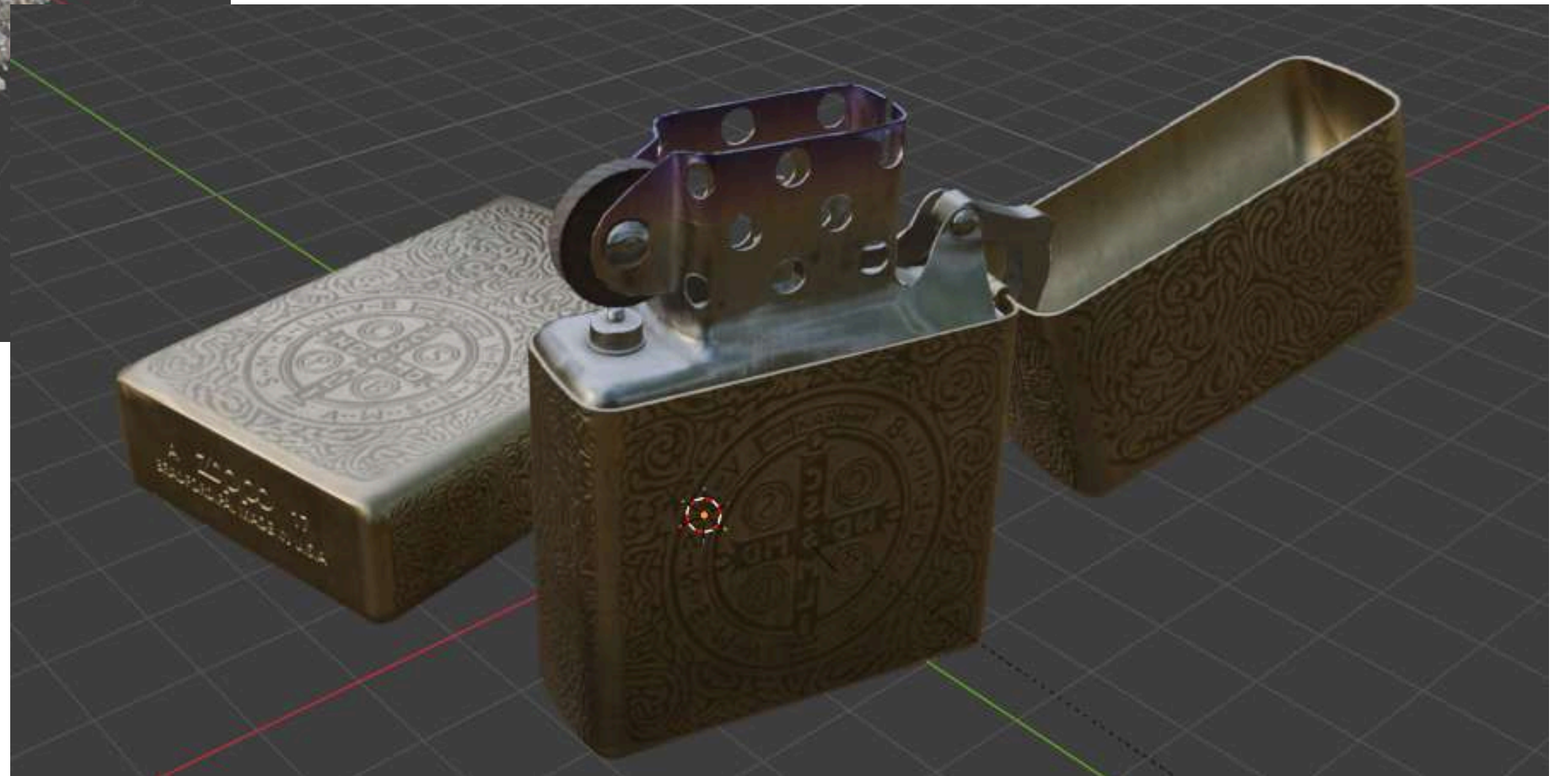
- Secondary camera renders the **destination island** into a texture on the portal plane
- Player viewpoint is **mirrored** through the frame so the preview matches what you see after walking through

MODELS (BLENDER)



Skeleton

Lighter



MODELS (THREE.JS)



Dock / Portal Frame

Shack / Hoe



TERRAIN



→ Expectation
No shader support in GLTF

→ Reality

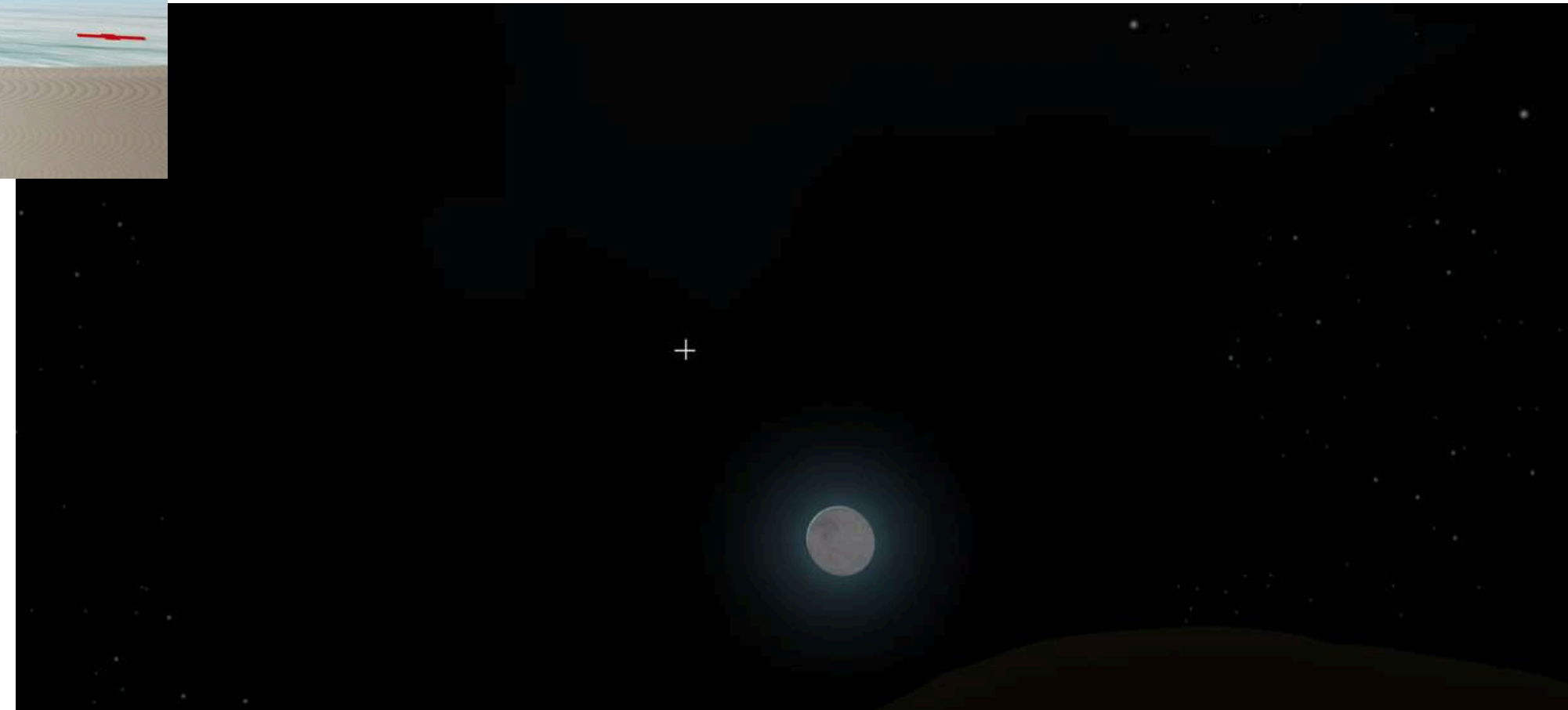


ILLUMINATION



- **Sun/Moon:** Directional lights on a 24-hour cycle; color and intensity adapt to time of day
- **Sky shader:** atmospheric scattering; sun&moon meshes + starfield at night
- **Atmospheric fog:** Exponential fog; tone and density evolve from day to night and per island

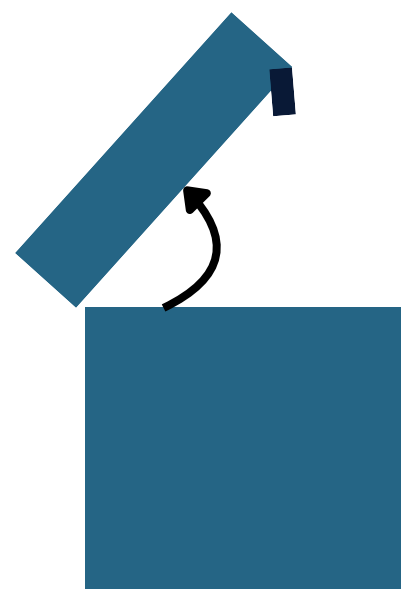
- **Real-time shadows:** PCF soft shadows
- **PBR materials & water:** Standard materials with texture maps; animated water shader for the ocean
- **Image finishing:** ACES tone mapping; selective bloom and SSAO for mood per island (AI)



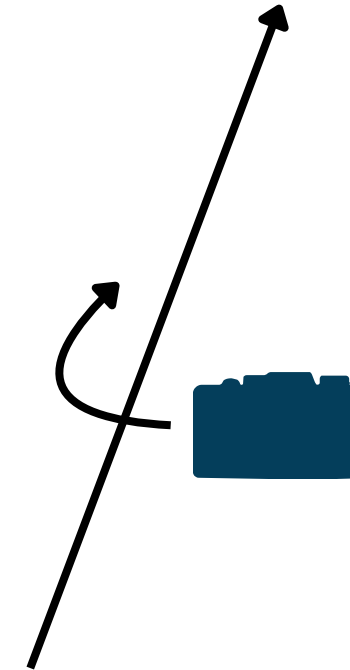
ANIMATION (BLENDER)



- Top - Relative to bottom
- Hinge - Relative to top

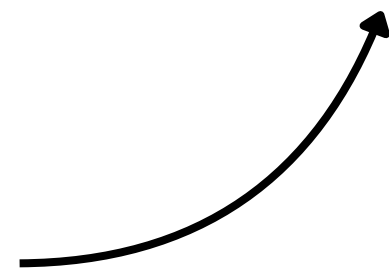


ANIMATION (THREE.JS)



→ Interaction triggered when a ray-cast hits the registred mesh

→ Exponencial fog



PERFORMANCE



- **Instance Mesh:** To load a ton of the same assets
- **GLTFLoader Wrapper:** Which cached the textures
- **Island Unloading:** Unloading already visited islands helped performance

DEVELOPMENT & AI

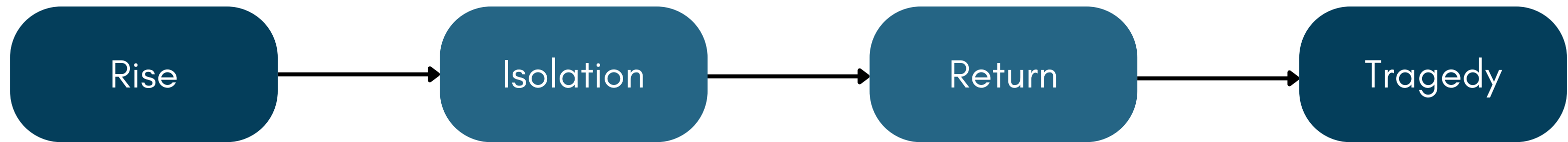
- Vite
- Typescript: Heavy focus on OOP
- Three.js (r183)
- Blender

Structure →

- Gemini: Research and learning Blender
- Claude: Game Logic

```
▼ ICG
  > .github
  ▼ public
    > hdri
    > island
    > models
  ▼ src
    > config
    > engine
    > interfaces
    > objects
    > platforms
    > player
    > puzzles
    > types
    > utils
    > world
  TS main.ts
```


FUTURE



Missing these 2 stages due to bugs and lack of work

→ Better PostFX

→ World scale / Building

→ Physics

REFERENCES

MODELS / TEXTURES

- SketchFab
- Polyhaven

PHYSICS

- Rapier.js

SHADERS

- Water shader - Three.js example

DEMO

ix-0.github.io/ICG/

THANK YOU.

ix-0.github.io/ICG/